

A Coincidence Algebra Bundle for Decay Quivers: An Algebraic Approach to Gamma-ray Spectroscopy*

Liam L. Schmidt[†]
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Motivated by the need for a more comprehensive algebraic structure to calculate coincidence probabilities of a general decay scheme for gamma ray spectroscopy, we model the decay scheme, rather naturally, as a quiver through which we define a *decay quiver*. The path algebra of quivers is the underlying, more general, algebra for transition matrices that is typically used in modeling decay schemes. The path algebra allows for concatenation of transitions which affords the calculation of cascade probabilities. We extend the path algebra to allow for the multiplication of non-composable paths, i.e., transition that don't directly share a level connecting them. We define the *coincidence algebra* as the algebra that allows for such an extension and realize it as the fibers for a *coincidence algebra bundle*, the base space of which is the path algebra where decay schemes live. A given decay schemes coincidence probabilities are calculated on its fiber. *Detection maps* are defined as maps on the base space that map transition probabilities to detection probabilities.

I. INTRODUCTION

The resultant gamma decay from nuclear decays such as β^+/β^- decay are often depicted by a decay diagram whose structure is analogous to those found in graph theory. Such a diagram is seen in Figure 1, where f_i are branching ratios to levels in the daughter nuclear and x_{ij} are the transition probabilities between level i and j . This work sets out to further generalize the treatment of these decay diagrams in an algebraic and geometric manner.

An algebraic structure for spectroscopic analysis is proposed that models the gamma decay as a quiver with a defined structure; a quiver is a directed graph upon which a vector space and algebra is composed [1]. It is common practice to treat nuclear levels using a graph theoretical approach. However, the manner in which that is done is through transition matrices [2, 3] and Markov processes [4], which take a matrix representation of the more fundamental algebra built from a quiver—the path algebra [1]. The path algebra constructed from a quiver that models a decay scheme allows for probability calculations for coincidence events (as do the transition matrices, though the path algebra keeps cascade terms separate and does not sum with the direct transitions). The conventional path algebra, however, is restricted to consecutive paths; that is, directly connected paths. In this work, the path algebra is extended into a tensor algebra where non-consecutive products may be calculated. This serves as a more general algebraic structure for nuclear level schemes, which may assist in the calculation of coincidence probabilities for gamma-ray spectroscopy.

A. Transition Matrices

Conventional treatment for decay diagrams is done with transition matrices [2]. General decays can be modeled using strictly lower triangular matrices, where the matrix algebra allows for the calculation of cascade prob-

abilities via powers of the transition matrix.

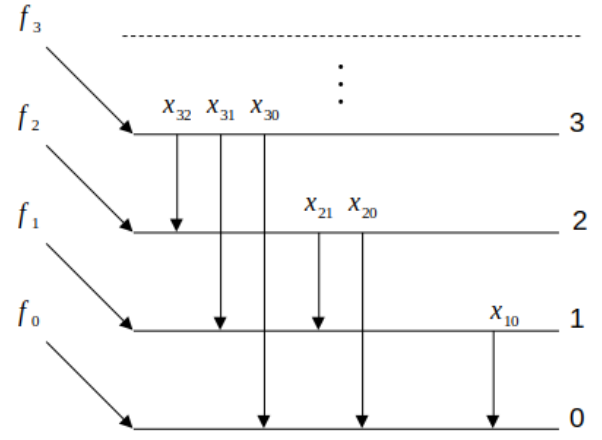


FIG. 1. General decay scheme with branching ratios f_i and transition probabilities x_{ij} .

For a general decay scheme, as shown in Figure 1, the transition probabilities can be stored in the transition matrix

$$x = \begin{pmatrix} 0 & & & & \\ x_{10} & 0 & & & \\ x_{20} & x_{21} & 0 & & \\ \vdots & \vdots & \vdots & \ddots & \\ x_{n0} & x_{n1} & x_{n2} & \cdots & x_{nn-1} & 0 \end{pmatrix}, \quad (1)$$

and the branching ratios are stored in a branching vector

$$f = (f_0, f_1, f_2, \dots, f_n). \quad (2)$$

Taking powers of the transition matrix will give cascade probabilities on the order of the power in the matrix position of the direct transition. For example, all two gamma

cascade probabilities can be calculated from:

$$x^2 = \begin{pmatrix} 0 & & \\ 0 & 0 & \\ x_{10}x_{21} & 0 & 0 \\ x_{10}x_{31} + x_{32}x_{20} & x_{21}x_{32} & 0 \end{pmatrix}. \quad (3)$$

To collect all orders of cascade probabilities, from single transitions to n gamma cascades, we may power expand the transition matrix:

$$\bar{x}^n = \sum_{k=0}^n x^k. \quad (4)$$

With the branching vector determining the level at which a given decay starts, the probability that a transition from level k to level l will occur for a given decay can be given as follows:

$$P_{TM}(x_{kl}) = [\text{diag}(f\bar{x}^n)x]_{kl}. \quad (5)$$

The diagonal matrix $\text{diag}(f\bar{x}^n)$ gives the probability that a given level is reached. To calculate the probability that two transitions, $k \rightarrow l$ and $q \rightarrow m$, occur, we take the probability of the first transition to occur multiplied by the probability that it leads to the second with

$$P_{TM}(x_{kl} \cap x_{qm}) = [\text{diag}([\bar{x}^n]_i)x]_{kl} [\text{diag}([\bar{x}^n]_{li})x]_{qm}. \quad (6)$$

The diagonal matrix $\text{diag}([\bar{x}^n]_{li})$ gives the probability that from level l the decay leads to a given level below it. The above probabilities are probabilities of emission; they indicate that the transition occurs. In gamma-ray spectroscopy we must deal with the probability of detection. It is the map between detected probabilities and emitted probabilities that is the practice of gamma-ray spectroscopy. To model this, the transition matrices can be mapped to include different detection probabilities, such as efficiencies. This is done in [2] where observed summing probabilities are given; this paper is motivated by [2] where we look to further generalize the mathematical procedure of coincidence probabilities. The work of Semkow et.al. [2] first established the transition matrix formalism for nuclear decay schemes, though largely building off the work of [5, 6]. In their paper, their motivation was to calculate the true coincidence summing corrections that occur in gamma-ray spectroscopy. When detecting the emission of gamma rays in a decay, there is some non-zero probability that two gammas from a cascade can hit your detector making it appear like the detection of a single gamma-ray with energy being the sum of the two. If the direct transition of that cascade exist, then extra counts are added to the peak of the direct transition and counts are lost from the the two smaller energy gamma peaks. When this occurs, we may refer to the loss of counts due to summing as summing out and the gain of counts as summing in. It is crucial in gamma-ray spectroscopy that the coincidence summing effect is well know in order to arrive at an accurate measurement. The transition matrix formalism was extended

to include different phenomena such as coincidence summing with X-rays [3]. The work of [7] takes a unique approach to modeling coincidence summing with hints of a graph theory, but falls short of reaching the rigorous formalism of quivers and path algebra. Recent work by Schmidt [8] gives a more comprehensive history of coincidence summing and also extends the matrix formalism of [2] for 180 degree coincidence (as a correction for coincidence summing) and for gated (coincidence) spectra. The work presented here is a further generalization of the transition matrix formalism for coincidence probabilities in nuclear level schemes.

The transition matrices are limited to creating product probabilities between connected transitions. This is due to the algebra of strictly lower triangular matrices. As shown in [2], this concatenation of probabilities allows for coincidence summing calculations of connected transitions. To calculate the coincidence summing of non-connected transitions, one must extend the algebra. It is the coincidence summing of non-connected transition in particular that motivates this work. To tackle this problem, we propose a structure that offers to deal with coincidence probabilities in the most general sense. The transition matrices emerge out of a more general algebra, called the path algebra of quivers. The transition matrices are isomorphic to the path algebra modded by the ideal of relations that equate all cascades to their direct transition. We extend this path algebra to a *coincidence algebra*.

II. QUIVERS AND PATH ALGEBRA

A quiver $\mathcal{Q} = (\mathcal{Q}_0, \mathcal{Q}_1, s, t)$ is a quadruple consisting of two sets: $\mathcal{Q}_0$ (whose elements are called vertices) and $\mathcal{Q}_1$ (whose elements are called arrows), and two maps $s, t : \mathcal{Q}_1 \rightarrow \mathcal{Q}_0$ which associate to each arrow $\alpha \in \mathcal{Q}_1$ its source $s(\alpha) \in \mathcal{Q}_0$ and its target $t(\alpha) \in \mathcal{Q}_0$, respectively. A *path* of length l is a sequence of connected arrows from $a \in \mathcal{Q}_0$ to $b \in \mathcal{Q}_0$ written as follows:

$$(a|\alpha_1, \dots, \alpha_l|b), \quad (7)$$

where $\alpha_i \in \mathcal{Q}_1$ is an arrow from vertex a . A path of length $l = 0$ is called the stationary path, denoted by

$$\epsilon_a = (a||a) \quad (8)$$

for vertex a . The path algebra $k\mathcal{Q}$ of $\mathcal{Q}$ is the k -algebra whose underlying k -vector space has as its basis the set $\mathcal{P}$ of all paths $(a|\alpha_1, \dots, \alpha_l|b)$ of length $l \geq 0$ in $\mathcal{Q}$ and such that the product of two basis vectors $(a|\alpha_1, \dots, \alpha_l|b)$ and $(c|\beta_1, \dots, \beta_k|d)$ of $k\mathcal{Q}$ is defined by

$$(a|\alpha_1, \dots, \alpha_l|b) \cdot (c|\beta_1, \dots, \beta_k|d) = \delta_{bc}(a|\alpha_1, \dots, \alpha_l, \beta_1, \dots, \beta_k|d). \quad (9)$$

The product of basis elements is extended to arbitrary elements of $k\mathcal{Q}$ by distributivity. For the sake of brevity, we use a hat notation to denote paths where for arbitrary

paths we write $\widehat{p}$ and for distinct paths the concatenation of arrows, $\alpha_1 \dots \alpha_l$, will be made explicit where the source and targets can be given by the s and t maps. The stationary paths are a complete set of idempotents for the algebra such that their sum gives the global identity element:

$$\mathbb{I}_{\mathcal{Q}} = \sum_{a \in \mathcal{Q}_0} \widehat{e}_a. \quad (10)$$

The vector space $k\mathcal{Q}$ has a direct sum decomposition into paths of different lengths:

$$k\mathcal{Q} = k\mathcal{Q}_0 \oplus k\mathcal{Q}_1 \oplus \dots \oplus k\mathcal{Q}_n \quad (11)$$

where $k\mathcal{Q}_l$ is the subspace whose basis $\mathcal{P}_l$ is the set of all paths of length l . The arrow ideal $R_{\mathcal{D}}$, which is defined as isomorphic to the quotient vector space $k\mathcal{Q}/k\mathcal{Q}_0$, may be decomposed into *source subspaces* $\mathcal{S}_i$ or *target subspaces* $\mathcal{T}_i$ composed of basis vectors with their source or target as level i , respectively:

$$R_{\mathcal{D}} = \mathcal{S}_1 \oplus \mathcal{S}_2 \oplus \dots \mathcal{S}_n, \quad (12)$$

$$R_{\mathcal{D}} = \mathcal{T}_0 \oplus \mathcal{T}_1 \oplus \dots \mathcal{T}_{n-1}. \quad (13)$$

We now define the following bilinear forms.

Definition II.1. For two paths $\widehat{p}_i, \widehat{p}_j \in P$, the *source bilinear form* $\langle, \rangle_s : k\mathcal{D} \rightarrow \mathbb{R}$ is given by

$$\langle \widehat{p}_i, \widehat{p}_j \rangle_s = \delta_{s(\widehat{p}_i)s(\widehat{p}_j)}. \quad (14)$$

Two paths $\widehat{p}_i, \widehat{p}_j$ are *diverging paths* if $\langle \widehat{p}_i, \widehat{p}_j \rangle_s = 1$

Definition II.2. For two paths $\widehat{p}_i, \widehat{p}_j \in P$, the *target bilinear form* $\langle, \rangle_t : k\mathcal{D} \rightarrow \mathbb{R}$ is given by

$$\langle \widehat{p}_i, \widehat{p}_j \rangle_t = \delta_{t(\widehat{p}_i)t(\widehat{p}_j)}. \quad (15)$$

Two paths $\widehat{p}_i, \widehat{p}_j$ are *converging paths* if $\langle \widehat{p}_i, \widehat{p}_j \rangle_t = 1$

Definition II.3. For two paths $\widehat{p}_i, \widehat{p}_j \in P$, the *path bilinear form* $\langle, \rangle_p : k\mathcal{D} \rightarrow \mathbb{R}$ is given by

$$\langle \widehat{p}_i, \widehat{p}_j \rangle_p = \delta_{s(\widehat{p}_i)s(\widehat{p}_j)} \delta_{t(\widehat{p}_i)t(\widehat{p}_j)}. \quad (16)$$

Two paths $\widehat{p}_i, \widehat{p}_j$ are *equivalent paths* if $\langle \widehat{p}_i, \widehat{p}_j \rangle_t = 1$

These bilinear forms extend to the entire vector space with distributivity. Projectors into the different subspaces can now be defined:

Definition II.4. The *source projector* $S_i : k\mathcal{Q} \rightarrow \mathcal{S}_i$ projects a vector $d \in k\mathcal{Q}$ into its source subspace $\mathcal{S}_i$:

$$S_i(d) = \sum_{\widehat{p} \in \mathcal{P}_1} \delta_{s(\widehat{p})i} \langle \widehat{p}, d \rangle_s \widehat{p}. \quad (17)$$

Definition II.5. The *target projector* $T_i : k\mathcal{Q} \rightarrow \mathcal{T}_i$ projects a vector $d \in k\mathcal{Q}$ into its target subspace $\mathcal{T}_i$:

$$T_i(d) = \sum_{\widehat{p} \in \mathcal{P}_1} \delta_{t(\widehat{p})i} \langle \widehat{p}, d \rangle_t \widehat{p}. \quad (18)$$

Definition II.6. The *target vertex projector* $V_t : k\mathcal{Q} \rightarrow k\mathcal{Q}_0$ projects a vector $d \in k\mathcal{D}$ into its idempotent subspace in a manner that sums over *converging* paths:

$$V_t(d) = \sum_{\widehat{e} \in \mathcal{P}_0} \langle \widehat{e}, d \rangle_t \widehat{e}. \quad (19)$$

Definition II.7. The *source vertex projector* $V_s : k\mathcal{Q} \rightarrow k\mathcal{Q}_0$ projects a vector $d \in k\mathcal{D}$ into its idempotent subspace in a manner that sums over *diverging* paths:

$$V_s(d) = \sum_{\widehat{e} \in \mathcal{P}_0} \langle \widehat{e}, d \rangle_s \widehat{e}. \quad (20)$$

Definition II.8. The *branching projector* $B : k\mathcal{Q} \rightarrow k\mathcal{Q}_0$ projects a decay vector $d \in k\mathcal{D}$ into its idempotent subspace $K\mathcal{D}_0$:

$$B(d) = \sum_{\widehat{e} \in \mathcal{P}_0} \langle \widehat{e}, d \rangle_p \widehat{e}. \quad (21)$$

The above definitions will aid in the following investigation in nuclear decay schemes formalized with quivers.

A. Decay Quiver

Definition II.9. A *decay quiver* is a finite connected acyclic quiver $\mathcal{D} = (\mathcal{D}_0, \mathcal{D}_1, s, t)$ where $\mathcal{D}_0$ is the set of vertices (levels), $\mathcal{D}_1$ is the set of arrows (transitions/gammas), and $s, t : \mathcal{D}_1 \rightarrow \mathcal{D}_0$ are maps that associate to each arrow in $\mathcal{D}_1$ its *source* and *target*, respectively.

We also define the following relation on $\mathcal{D}_0$:

Definition II.10. Let $N : \mathcal{D}_0 \rightarrow \mathbb{N}$ and $\widehat{e} \in \mathcal{D}_0$ where

$$N(\widehat{e}) = \sum_{\widehat{p} \in \mathcal{P}_1} \langle \widehat{e}, \widehat{p} \rangle_s, \quad (22)$$

that is, it returns the number of diverging arrows for that vertex. A relation $\preceq$ can be established on the vertex set that turns it into a totally ordered set defined by the following:

$$\widehat{e}_a \preceq \widehat{e}_b \Leftrightarrow N(\widehat{e}_a) \leq N(\widehat{e}_b). \quad (23)$$

Proposition II.1. A decay quiver with n levels is fully connected if a total ordering can be defined on the set of vertices $\mathcal{D}_0$ via the relation $\preceq$ with a bijective map $N : \mathcal{D}_0 \rightarrow \mathbb{N}_{<n}$.

In all further derivations and discussion, we take any decay quiver with n levels to be fully connected unless otherwise stated. The bijection between $\mathcal{D}_0$ and $\{0, 1, 2, \dots, n-1\}$ gives the number of arrows in the quiver, that is, the cardinality of the arrow set, to be $|\mathcal{D}| = n(n-1)/2$. Any given decay vector $d \in K\mathcal{D}$ can be decomposed into a sum of its source vectors (or target vectors) d_i , which are the vectors of the source(target)

projectors, and its branching vector, which is the projected vector of stationary paths:

$$d = B(d) + \sum_{i=0}^n S_i(d) = b + \tau. \quad (24)$$

The source vectors $S_i(d)$ and the branching vector b are then unit vectors under the l_1 -norm $\|\cdot\|$, given the probability conditions of the level transitions and the branching. Decay vectors may thus be classified as unit vectors under a *decay norm* $\|d\|_{\mathcal{D}}$ defined as

$$\|d\|_{\mathcal{D}} = \frac{1}{n} \sum_{j < i}^n |x_{ij}| = 1, \quad (25)$$

which is a sum over all the components of the decay vector divided by the number of levels. This norm further defines the subset in which the decay vectors live in; that is, in the subspace $k\mathcal{D}_0 \oplus k\mathcal{D}_1$ with the constraint $\|d\|_{\mathcal{D}} = 1$. Analogous to the probabilities given in equations 5 and 6, we can formulate the probability of a given transition in the path algebra using the projectors we defined:

$$P(\hat{p}_i) = \langle V_t(b \cdot \bar{\tau}^n) \cdot \tau, \hat{p}_i \rangle_p, \quad (26)$$

along with the coincidence probability,

$$P(\hat{p}_i \cap \hat{p}_j) = \langle V_t(b \cdot \bar{\tau}^n) \cdot \tau, \hat{p}_i \rangle_p \langle V_t(\epsilon_{t(\hat{p}_i)} \cdot \bar{\tau}^n) \cdot \tau, \hat{p}_j \rangle_p, \quad (27)$$

where $\tau = d - b$ as defined in Eq. 24, that is, the decay vector with the branching terms subtracted off. In this formulation, the target vertex projector acts as the diagonalization (left multiplying) of the decay vector with the branching from above terms multiplied through. To select a given transition the path equivalence bilinear form is used. It should be noted, given that the target vertex projector gives back a vector of stationary paths, when multiplied onto the decay vector d , what is given is a vector in $k\mathcal{D}_1$ still. Therefore, when finding a given path with the path form only single terms are produced—no equivalence between different path lengths is produced yet, as would be given by the path form in general. To arrive at notions of coincidence summing, which in this formalism we may generally state as the path equivalence between paths of different lengths, one must power expand the decay vector which V_t is being multiplied onto to create the higher order path terms. Algebraically, one may think of coincidence summing as path equivalence; however, a more complete depiction of coincidence summing is more aptly posed in the context of detection maps.

B. Detection Maps

For a given decay quiver, a decay vector living in the path algebra models a nuclear level scheme with n levels.

The decay vectors components give the probability for the transition corresponding to the basis vector path. As such, the decay vectors are units vectors under the decay norm. However, there of course exist vectors in the path algebra that do not correspond to transition probabilities satisfying the norm condition, but may still be considered as probabilities. In this manner, detection vectors may also live in the path algebra where the components not only include the transition probabilities but also efficiencies or different detection probabilities. We posit then that the path algebra for a given quiver is the space of all detection nuclear level schemes, such that transition probabilities are mapped to detection probabilities in this space. If we compute probabilities via Eq. 26 with a detection mapped decay vector, where efficiencies are now multiplied onto the transition probabilities, what we get is not the probability of emission but the probability of detecting that gamma/transition. In general, for more realistic observations of decaying nuclei, there are other factors beside detector efficiency that go into the detected probabilities that one doing gamma spectroscopy must consider when attempting to map their way back to the transition probabilities; one can consider a number of different detection maps that take the decay vectors describing nuclear decay schemes into different vectors that account for different probabilities involved in detection. A detection map that corresponds to the full energy hit of a gamma may be given by,

$$\varphi_h(\tau) = \sum_{\hat{p} \in \mathcal{P}_1} e_{\hat{p}}^p \langle \hat{p}, \tau \rangle_p \hat{p}, \quad (28)$$

where $e_{\hat{p}}^p$ is the peak efficiency for the gamma associated to the path $\hat{p}$ of length one. We may also consider the avoidance of a gamma at any energy (full detection or Compton scattered). To do so, however, we must consider the detector geometry. When talking about gammas summing together and avoiding summing out one must consider angular correlations: the probability that two gammas are emitted at a certain angle. For now, we just consider isotropic emission such that the probability for two gammas hitting the same detector is just given by the number of detectors. Therefore, we may map a decay vector to a detection vector that accounts for the probability of it hitting a given detector at any energy:

$$\varphi_a(\tau) = \sum_{\hat{p} \in \mathcal{P}_1} \frac{e_{\hat{p}}^t}{N} \langle \hat{p}, \tau \rangle_p \hat{p}, \quad (29)$$

for N detectors (one may imagine N detectors spherically placed around an emission source) where $e_{\hat{p}}^t$ is the total efficiency. To arrive at a notion of summing out, we consider the probability that a given gamma does *not* hit a given detector at any energy:

$$\varphi_o(\tau) = \sum_{\hat{p} \in \mathcal{P}_1} \left[1 - \frac{e_{\hat{p}}^t}{N} \right] \langle \hat{p}, \tau \rangle_p \hat{p}, \quad (30)$$

These are all maps analogous to those defined in [2] with the transition matrix formalism and used in the literature afterwards to discuss summing corrections. To account for the summing in, cascade probabilities are built via the power expansion of the decay vector. However, if the power expansion creates the gammas of interest with cascade terms to account for summing in, we must now consider the detector geometry involved for the summing in. Maintaining our isotropic approximation, summing in probabilities can just be considered with the number of detectors ([2] considers only one detector so the efficiencies are sufficient in accounting for summing probabilities. For multiple detectors one must extend this of course). We then redefine the power expansion used for detection mapped decay vectors with isotropic emission.

$$\overline{\varphi(\tau)}_N^n = \sum_{k=0}^n \frac{\tau^k}{N^{k+1}}, \quad (31)$$

with k the degree of summing. We may now give a path algebraic formulation for the probability vector that provides the probability of detection for a given transition where summing in and summing out effects are considered:¹.

$$\Gamma(d) = V_t \left(b \cdot \overline{\varphi_o(\tau)}^n \right) \cdot \left[\overline{\varphi_h(\tau)}_N^n - \mathbb{I}_Q \right] \cdot V_s \left(\overline{\varphi_o(\tau)}^n \right). \quad (32)$$

In contrast from equation 26, the target vertex projector now contains probabilities multiplied onto vertices (stationary paths) that account for summing out from above, $\overline{\varphi_h(\tau)}_N^n$ contains the paths with summing in given via the power expansion, and the source vertex projector now gives summing out from below. All of which, when given together as shown above, provide a vector where the probability of detecting a transition is given by the component of the corresponding basis vector/path. However, different from the transition matrix formalism where the summing in effects are added with the direct transition, the summing in terms exist in separate paths, i.e. they have their own basis vector. To add them back together, the path equivalence form, which equates paths with the same source and target—but does not distinguish between path length, can be used to calculate probabilities for detecting a given transition with summing effects accounted for:

$$P_{PA}(\hat{p}_i) = \langle \Gamma(d), \hat{p}_i \rangle_p. \quad (33)$$

Up to here, this new formulation just restates the transition matrix formalism with the caveat that summing in terms are kept separate; the path equivalent quotient path algebra, that is, the path algebra with the path equivalence modded out, is isomorphic to the transition

matrices. However, this new formalism may now be extended beyond the path concatenation that we are restricted to with the transition matrices and also allow for a greater separability between equivalent paths. If we wish to allow for not only paths – being connected arrows – but pairs, or even sets, of non-connected arrows (i.e., coincidences in general) in the algebra (as shown in Figure 2), we must extend the algebra to a tensor decay space defined in the following section

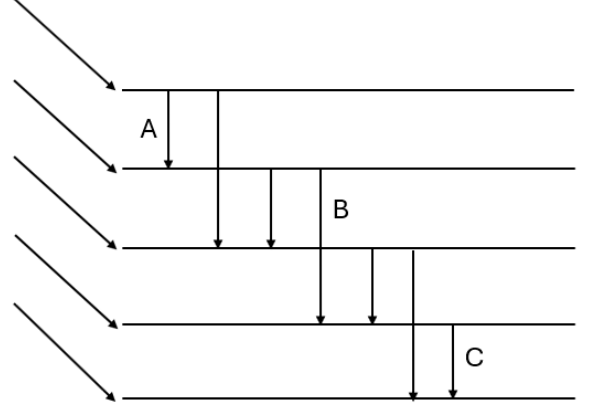


FIG. 2. General decay diagram showing the connected transitions A and B and the non-connected or *coincidence* transitions A and C.

III. COINCIDENCE ALGEBRA BUNDLE

A. Coincidence Tensor Space

The path algebra vector space can be extended to include basis vectors that afford product probabilities between non-connected paths by taking its tensor product expansion. The *tensor decay space* $\mathcal{T}$ can be defined as follows:

$$\mathcal{T} = k\mathcal{D} \oplus (k\mathcal{D} \otimes k\mathcal{D}) \oplus \dots \oplus (k\mathcal{D}^{\otimes n}) = \bigoplus_{i=0}^n k\mathcal{D}^{\otimes i} \quad (34)$$

This larger space contains – as basis vectors – all combinations of paths in the form of tensor products; not only non-connected but overlapping paths as well. However, it is not this total space that we require for our coincidences. In fact, only the physically realized coincidences are needed—these being the pairs of arrows that can be observed in a single decay event. For this reason, we consider a subset of this tensor space:

Definition III.1. Let $\mathbb{P}(\mathcal{P}_0 \cup \mathcal{P}_1)$ be the power set of $\mathcal{P}_0 \cup \mathcal{P}_1$, that is, the set of all possible subsets of $\mathcal{P}_0 \cup \mathcal{P}_1$. The coincidence set $\mathcal{O}$ is a subset of $\mathbb{P}(\mathcal{P}_0 \cup \mathcal{P}_1)$ and is defined as

$$\mathcal{O} \subset \mathbb{P}(\mathcal{P}_0 \cup \mathcal{P}_1) = \left\{ u \in \mathbb{P}(\mathcal{P}_0 \cup \mathcal{P}_1) \mid |u| \geq 2, \forall \hat{p}_1, \hat{p}_2 \in u, s(\hat{p}_2) \preceq t(\hat{p}_1) \vee s(\hat{p}_1) \preceq t(\hat{p}_2) \right\}, \quad (35)$$

where the paths in a given element of $\mathbb{P}(\mathcal{P}_0 \cup \mathcal{P}_1)$ with cardinality greater than or equal to two do not overlap.

The coincidence set may now be used to define the coincidence space.

Definition III.2. Let $o \in \mathcal{O}$ be totally ordered such that $\forall \hat{p}_1, \hat{p}_2 \in o, \hat{p}_1 < \hat{p}_2 \iff N(s(\hat{p}_2)) < N(s(\hat{p}_1))$; that is, it is ordered by the higher paths in the quiver, and let o be indexed by this ordering. The *coincidence vector space* $\mathcal{C}$ is defined as

$$\mathcal{C} := \text{span} \left\{ t = \bigotimes_{i=1}^{|o|} \hat{p}_i \mid \hat{p}_i \in o, \forall o \in \mathcal{O} \cup \mathcal{P}_0 \cup \mathcal{P}_1 \right\} / \mathcal{I}. \quad (36)$$

Where $\mathcal{I}$ is an identity ideal of relations which allow for the stationary paths to act as local identities still:

$$\mathcal{I} = \langle s(\hat{p}_i) \otimes \hat{p}_i - \hat{p}_i, \hat{p}_i \otimes t(\hat{p}_i) - \hat{p}_i \mid \forall \hat{p}_i \in \mathcal{P}_1 \rangle. \quad (37)$$

We include $\mathcal{P}_0 \cup \mathcal{P}_1$ to retain the single transition that were exclude in the definition of $\mathcal{O}$. The basis elements of $\mathcal{C}$ are called *coincidences*.

Definition III.3. The coincidence vector space $\mathcal{C}$ is graded by the number of transitions in a tensor product. That is,

$$\mathcal{C} = \bigoplus_{i=0}^n \mathcal{C}_i, \quad (38)$$

where $\mathcal{C}_i$ is the subspace spanned by basis elements containing exactly i transition paths, with stationary paths assigned degree zero.²

Definition III.4. The source and target maps defined in the path algebra extend to coincidences (tensor products) as follows:

$$s(\hat{p}_i \otimes \hat{p}_j \otimes \dots \otimes \hat{p}_n) = s(\hat{p}_i) \quad (39)$$

$$t(\hat{p}_i \otimes \hat{p}_j \otimes \dots \otimes \hat{p}_n) = s(\hat{p}_n) \quad (40)$$

B. Coincidence Algebra Bundle

With the coincidence decay space now defined, we may consider an algebra on this vector space. The path algebra defined for a quiver uses concatenation of paths. In this manner, a given vector, when squared, will acquire components of concatenated paths where the coefficients are multiplied (i.e., it is a graded algebra). In the context of decays this alludes to coincidence probabilities or joint probabilities of the two paths. When considering non-connected pairs/sets-of-transitions, i.e. coincidences, one must consider what type of multiplication is appropriate when building them out of transitions. In the event that the coincidences contain directly connected transitions, it must also reduce to the original path algebra. When considering coincidences and their probabilities, what matters is precisely how they are not connected. A degree of connectivity can be defined by a sum over connecting paths. This is made clear by considering two gammas in the decay. If one gamma is at a higher level, what determines if that decay will lead to the second gamma is the sum of the probabilities contained in the different paths that connect them. This can be seen in Figure 2 where gamma A and C are only partially connected given that there exist a decay from A to the ground state that does not emit gamma C. The connectivity between A and C is the probability that the tail vertex of A leads to the source vertex of C given the intermediary transitions.

If the two gammas are *directly connected*, i.e., concatenate to a larger path, then the sum will simply reduce to one. In the path algebra formalism, the decay scheme exist as a decay vector in the decay vector space; we simply get correct product of probabilities since the transition probabilities are contained in the components of the decay vector, i.e., they are inherently in the vector multiplication by use of the decay vector. When a decay vector existing in the decay vector space – which is now taken to be a subspace of the coincidence decay space – is squared, the product needs to allow for coincidence terms where the coefficients have the appropriate values; i.e., the sum of the paths connecting them. The vector multiplication in the coincidence decay space therefore must depend on the decay vector; i.e., the product of two basis vectors in the coincidence space needs to give a third vector that has a scalar factor which is a function of the decay vector. An single algebra therefore can not be defined on the entire decay vector space and what is needed is a family of algebras. In particular, what we arrive at is an *algebra bundle*, i.e., a fiber bundle where the base space is the path algebra of the decay quiver and the fibers are a coincidence algebra defined at every point in the base space. Before we define the algebra,

¹ This is analogous to the matrix $S = NAM$ given in [2] with the rate R taken out

² Note, the grading on $\mathcal{C}$ is not the same grading that is on the tensor space $\mathcal{T}$ as tensor degree grading is broken by the quotient with the identity ideal. However, with grading defined by degree of coincidence with stationary paths degree zero and transitions degree one, the grading of the coincidence space does follow.

let us start with the notion of connectivity between two paths.

Definition III.5. The *scalar connection* $c_d : k\mathcal{D} \times k\mathcal{D} \rightarrow \mathbb{R}$ between two paths $\hat{p}_i, \hat{p}_j \in \mathcal{P}$ in a decay vector d is defined as follows:

$$c_d(\hat{p}_i, \hat{p}_j) = \langle V_t(t(\hat{p}_i) \cdot \bar{\tau}^n), s(\hat{p}_j) \rangle_p. \quad (41)$$

Similar to the path algebra in Eq. (9), the product of basis elements (paths in $\mathcal{P}$), extends to arbitrary elements of $k\mathcal{D}$ by distributivity, allowing the scalar connection to be defined on $k\mathcal{D} \times k\mathcal{D}$.

Lemma III.1. The scalar connection satisfies the 2-cocycle condition

$$c_d(\hat{p}_j, \hat{p}_k) c_d(\hat{p}_i, \hat{p}_j \otimes \hat{p}_k) = c_d(\hat{p}_i, \hat{p}_j) c_d(\hat{p}_i \otimes \hat{p}_j, \hat{p}_k). \quad (42)$$

Proof. By Definition III.4, the scalar connection only depends on the source and target maps of the paths/coincidences. Consequently,

$$c_d(\hat{p}_i, \hat{p}_j \otimes \hat{p}_k) = c_d(\hat{p}_i, \hat{p}_j), \quad (43)$$

likewise

$$c_d(\hat{p}_i \otimes \hat{p}_j, \hat{p}_k) = c_d(\hat{p}_j, \hat{p}_k). \quad (44)$$

Thus, both sides of Eq. 41 reduce to $c_d(\hat{p}_i, \hat{p}_j) c_d(\hat{p}_j, \hat{p}_k)$. $\square$

As previously mentioned, when the paths are connected by a single vertex, the scalar connection just returns a value of one. In separated cases, the scalar connection gives a sum over paths with the weights given by the decay vector. In the meaning of the decay, the scalar connection gives the probability that the higher transition will lead to the lower transition. If the scalar connection is one, the two paths are *directly connected*. If the scalar connection is less than one, the two paths are *partially connected*. With the scalar connection, we may now define the coincidence algebra.

Definition III.6. Let $d \in k\mathcal{D}$ be a decay vector. The coincidence algebra $\mathcal{C}_d$ is the vector space $\mathcal{C}$ equipped with the bilinear multiplication $\cdot_d$ defined on basis elements by

$$\hat{p}_i \cdot_d \hat{p}_j = c_d(\hat{p}_i, \hat{p}_j) \hat{p}_i \otimes \hat{p}_j, \quad (45)$$

for $\hat{p}_i, \hat{p}_j \in k\mathcal{D}$, and extended linearly to $\mathcal{C}$ via Definition III.4.

Proposition III.1. The multiplication $\cdot_d$ defined in Definition III.6 is closed and associative. Hence, $\mathcal{C}_d$ is an algebra.

Proof. Closure follows given that the product of every basis element either give a coincidence basis element following the grading in Definition III.3 or is zero due to ordering constraints of transitions shown in Definition III.1. Associativity is shown as follows. Given $\hat{p}_i, \hat{p}_j, \hat{p}_k \in \mathcal{C}$,

$$\begin{aligned} (\hat{p}_i \cdot_d \hat{p}_j) \cdot_d \hat{p}_k &= c_d(\hat{p}_i, \hat{p}_j) \hat{p}_i \otimes \hat{p}_j \cdot_d \hat{p}_k \\ &= c_d(\hat{p}_i, \hat{p}_j) c_d(\hat{p}_i \otimes \hat{p}_j, \hat{p}_k) \hat{p}_i \otimes \hat{p}_j \otimes \hat{p}_k \end{aligned} \quad (46)$$

and

$$\begin{aligned} \hat{p}_i \cdot_d (\hat{p}_j \cdot_d \hat{p}_k) &= c_d(\hat{p}_j, \hat{p}_k) \hat{p}_i \cdot_d \hat{p}_j \otimes \hat{p}_k \\ &= c_d(\hat{p}_j, \hat{p}_k) c_d(\hat{p}_i, \hat{p}_j \otimes \hat{p}_k) \hat{p}_i \otimes \hat{p}_j \otimes \hat{p}_k. \end{aligned} \quad (47)$$

Where the above equations are made equal by Lemma III.1. $\square$

Moving forward, we must consider consistent decay structures in the decay vector space. If a decay vector space is constructed from a fully connected decay quiver, then the space will allow for all possible transitions. However, for a given gamma decay scheme, it is usually not the case that it can be represented by a fully connected decay quiver. Rather, its decay structure exist in some subspace of the fully connected decay vector space where forbidden transitions are dimensions excluded in the subspace describing the decay structure. We must recognize this because when comparing coincidence algebras at different decay vectors $d \in k\mathcal{D}$, if the decay vectors are of a fundamentally different decay structure, that is, one includes a transition that is forbidden in the other, then the map from one decay vector to another may give a scalar connection of zero—not due to failure of transitions being properly ordered, but due to transitions probabilities being equal to zero. If we wish to establish any equivalence between coincidence algebras at different decay vectors, we must restrict the equivalence to structurally consistent decay schemes, which we define as follows.

Definition III.7. Let

$$d = \sum_{p \in \mathcal{P}_0 \cup \mathcal{P}_1} x_p \hat{p}$$

be a decay vector. The support of d is defined by

$$\text{supp}(d) = \{p \in \mathcal{P}_0 \cup \mathcal{P}_1 \mid d_p \neq 0\}.$$

Two decay vectors $d, d' \in k\mathcal{D}$ are said to have the same decay structure, written

$$d \sim d',$$

if

$$\text{supp}(d) = \text{supp}(d').$$

The corresponding consistent decay region associated with d is the equivalence class

$$B_d = [d] = \{d' \in k\mathcal{D} \mid d' \sim d\}. \quad (48)$$

Lemma III.2. For two coincidence algebras $\mathcal{C}_d$ and $\mathcal{C}_{d_0}$ with equivalent decay structure such that $d \sim d_0$, there exist a family of isomorphisms $F_d : \mathcal{C}_d \rightarrow \mathcal{C}_{d_0}$ where

$$F_d(c) = \sum_{q=1}^m a_q \lambda_{d_0}(\hat{c}_q) \hat{c}_q \quad (49)$$

where $c \in \mathcal{C}_d$ with basis elements $\hat{c}_q$ and $\lambda_{d_0}(\hat{c}_q)$ is a scaling of the basis element

Proof. Two coincidences algebras C_d and C_{d_0} are algebra isomorphisms if

$$F_d(\widehat{p}_i \cdot_d \widehat{p}_j) = F_d(\widehat{p}_i) \cdot_{d_0} F_d(\widehat{p}_j). \quad (50)$$

which extends to the entire coincidence space. The LHS gives

$$F_d(\widehat{p}_i \cdot_d \widehat{p}_j) = \lambda_{d_0}(\widehat{p}_i \otimes \widehat{p}_j) c_{d_0}(\widehat{p}_i, \widehat{p}_j) \widehat{p}_i \otimes \widehat{p}_j, \quad (51)$$

and the RHS gives

$$F_d(\widehat{p}_i) \cdot_{d_0} F_d(\widehat{p}_j) = \lambda_{d_0}(\widehat{p}_i) \lambda_{d_0}(\widehat{p}_j) c_{d_0}(\widehat{p}_i, \widehat{p}_j) \widehat{p}_i \otimes \widehat{p}_j \quad (52)$$

The scalars in the isomorphism may be determined by following the graded structure of the coincidence space. Letting the scalars of coincidence degree one (transitions), $\lambda_{d_0}(\widehat{p}_i)$ and λ_{d_0} be free, the scalars for higher degree coincidences may be determined.

$$\lambda_{d_0}(\widehat{p}_i \otimes \widehat{p}_j) = \lambda_{d_0}(\widehat{p}_i) \lambda_{d_0}(\widehat{p}_j) \frac{c_{d_0}(\widehat{p}_i, \widehat{p}_j)}{c_d(\widehat{p}_i, \widehat{p}_j)} \quad (53)$$

Using Lemma III.1 and by setting the free transition scaling parameters to one, we may extend to higher degree coincidence:

$$\lambda_{d_0}(\widehat{p}_i \otimes \widehat{p}_j \otimes \dots \otimes \widehat{p}_q \otimes \widehat{p}_n) = \frac{c_{d_0}(\widehat{p}_i, \widehat{p}_j) \dots c_{d_0}(\widehat{p}_q, \widehat{p}_n)}{c_d(\widehat{p}_i, \widehat{p}_j) \dots c_d(\widehat{p}_q, \widehat{p}_n)} \quad (54)$$

That is, for structurally equivalent coincidence algebras, since variation in the algebra is only induced via a change in the scalar connection that keeps it positive, an isomorphism can be established by canceling out this scaling from the scalar connection with a λ determined for each coincidence element in $\mathcal{C}$. Therefore all structurally equivalent coincidence algebras are isomorphic. $\square$

Theorem III.3. The disjoint union of all structurally equivalent coincidence vector spaces C_d for $d \in B_{d_0}$ denoted,

$$\mathcal{A} = \bigsqcup_{d \in B_{d_0}} C_d \quad (55)$$

where d_0 is a fixed decay vector admits a trivial algebra bundle $\pi : \mathcal{A} \rightarrow B_d$ up to isomorphisms $\{F_d\}_{d \in B_{d_0}}$ such that

$$\mathcal{A} \cong B_{d_0} \times C_{d_0}. \quad (56)$$

with $\pi^{-1}(d) = C_d$

Proof. $\mathcal{A}$ is a trivial algebra bundle if a global trivialization Ψ can be found such that

$$\Psi : \mathcal{A} \rightarrow B_{d_0} \times C_{d_0}. \quad (57)$$

The family of isomorphisms $\{F_d\}_{d \in B_{d_0}}$ may be used to complete this mapping with

$$\Psi(c) = (\pi(c), F_{\pi(c)}(c)) \quad (58)$$

for $c \in C_d$. That is, a global trivialization can be found that is just the rescaling of a fixed product bundle $B_{d_0} \times C_{d_0}$ via the isomorphism F_d for all $d \in B_{d_0}$. $\square$

When considering a decays vectors coincidence probabilities that are calculated in its fiber, we require sections that lift that decay vector into its fiber in a manner that copies the vector into the space of the fiber – which is always allowed given $k\mathcal{D} \subset \mathcal{C}$. We then denote the section $\tilde{s} : B_{d_0} \rightarrow \mathcal{A}$ such that $d \mapsto (d, \tilde{s}(d))$. The underlying vector bundle of the algebra bundle is a product bundle $B_{d_0} \times \mathcal{C}$, thus elements of the bundle are tuples (a, b) with $a \in B_{d_0}$ and $b \in \mathcal{C}$. However, when considering a vector in the bundle, the base space vector is what defines the algebra; what is being multiplied is the vector in $\mathcal{C}$, i.e., multiplying two vectors in a fiber defined at point $a \in B_{d_0}$ is defined as

$$b \cdot_a c = (a, b)(a, c) = (a, b \cdot c). \quad (59)$$

With this in mind, further notation restricts to writing the coincidence vector only with the path vector defined in the product. With the connection between levels and transitions now defined within the algebra, the probability vector of a given transition $\Gamma_1(d)$ and for two transitions in coincidence $\Gamma_2(d)$ can be given by

$$\Gamma_1(d) = \tilde{b} \cdot_d \tilde{\tau}, \quad (60)$$

$$\Gamma_2(d) = \tilde{b} \cdot_d \tilde{\tau} \cdot_d \tilde{\tau}. \quad (61)$$

Where $\tilde{b} = (d, b(d))$ —the branching section evaluated at d , is the branching vector in the fiber at d and $\tilde{\tau} = (d, \tau(d))$ —the transition section evaluated at d , is the transition vector in the same fiber, where $b(d)$ and $\tau(d)$ are defined in Equation 24. As with the path algebra, the order of multiplication is important and must be ordered in the hierarchy of the quiver, that is, the first term contains the branching ratios on the stationary paths which is connected to all paths below it via the scalar connection defined in the coincidence algebra. To get the coincidence probabilities we then further connect it to all paths below it; this is done by simply *coincidence multiplying* onto the (lifted) transition vector again. What is different now from the path algebra formulation is that we now have terms in the probability vector that correspond to coincidences of paths that are not directly connected; e.g., with this a term for the coincidence of x_{32} and x_{10} is now given. Not only that, but for each transition, the level at which it decayed from is now a separate basis vector as the transition is now tensored with the vertices. Examples for $\Gamma_1(d)$ and $\Gamma_2(d)$ for a general decay scheme with $n = 3$ is shown in Tables I and II, respectfully. We also point out how easily gated probability vectors emerge out of the coincidence algebra—indeed, that is its strength. The transition matrix formalism allows for the summing corrections for all of the *singles* spectra (single gamma-ray peaks) in the form of a matrix. As shown in equation 6, the coincidence probability for two *specific* gamma can be given; however, there is no equation the produces all possible coincidences (such would require a rank for tensor, with

Basis Vector	Branch f_3	Branch f_2	Branch f_1
$\widehat{p}_{30}$	$f_3 x_{30}$		
$\widehat{p}_{31}$	$f_3 x_{31}$		
$\widehat{p}_{32}$	$f_3 x_{32}$		
$\widehat{e}_3 \otimes \widehat{p}_{21}$	$f_3 x_{32} x_{21}$		
$\widehat{e}_3 \otimes \widehat{p}_{20}$	$f_3 x_{32} x_{20}$		
$\widehat{e}_3 \otimes \widehat{p}_{10}$	$f_3 (x_{31} + x_{32} x_{21}) x_{10}$		
$\widehat{p}_{20}$		$f_2 x_{20}$	
$\widehat{e}_2 \otimes \widehat{p}_{10}$		$f_2 x_{21} x_{10}$	
$\widehat{p}_{10}$			$f_1 x_{10}$

TABLE I. Probability vector $\Gamma_1(d)$ coefficients for a 4 level decay quiver. The basis vectors determine the transition and the coefficients give the probability of that transition for a given decay. The basis vectors are separated by the branch they decayed from.

Basis Vector	Branch f_3	Branch f_2
$\widehat{p}_{31} \otimes \widehat{p}_{10}$	$f_3 x_{31} x_{10}$	
$\widehat{p}_{32} \otimes \widehat{p}_{20}$	$f_3 x_{32} x_{20}$	
$\widehat{p}_{32} \otimes \widehat{p}_{10}$	$f_3 x_{32} x_{21} x_{10}$	
$\widehat{p}_{32} \otimes \widehat{p}_{21}$	$f_3 x_{32} x_{21}$	
$\widehat{e}_3 \otimes \widehat{p}_{21} \otimes \widehat{p}_{10}$	$f_3 x_{32} x_{21} x_{10}$	
$\widehat{p}_{21} \otimes \widehat{p}_{10}$		$f_2 x_{21} x_{10}$

TABLE II. Probability vector $\Gamma_2(d)$ coefficients for a 4 level decay quiver. The basis vectors determine the coincidence and the coefficients give the probability of that coincidence for a given decay. The basis vectors are separated by the branch they decayed from.

two indices for each gamma). In [8], they showed how to calculate the final probability matrix with summing corrections for a given gate (gamma in coincidence) by partitioning the transition matrix into block matrices and treating the block matrices with the Semkow formalism separately before direct summing them back together. In the coincidence algebra, all possible coincidences can be given by single vector. To gate on a certain gamma, $\Gamma_C(d)$ can be projected onto a subspace defined by the gate. The partitioned Semkow formalism in [8] indeed gives the transition matrix representation of $\Gamma_C(d)$ projected onto a gated subspace. Projections of coincidence vectors can be defined in the following manner. First, we begin by formulating how to compare tensor products of paths. This can be done by defining forms that sum over the paths in a tensor product:

$$[\widehat{p}_i, \widehat{p}_1 \otimes \widehat{p}_2 \otimes \dots \otimes \widehat{p}_j]_p = \sum_{k=1}^j \langle \widehat{p}_k, \widehat{p}_i \rangle_p, \quad (62)$$

which expands linearly for a given vector $c \in \mathcal{C}$ by

$$[\widehat{p}_i, c]_p = \sum_{q=1}^m a_q [\widehat{p}_i, \widehat{c}_q]_p, \quad (63)$$

where

$$c = \sum_{q=1}^m a_q \widehat{c}_q, \quad (64)$$

with $\widehat{c}_q$ denoting some tensor product of paths (a basis vector in $\mathcal{C}$). We therefore define the following gate projector

Definition III.8. The gate projector $G_1(\widehat{p}_i, \widehat{c}) : \mathcal{C} \rightarrow \mathcal{C}$ projects a vector $c \in \mathcal{C}$ onto the subspace whose basis vectors overlap with a path $\widehat{p}_i$ such that $[\widehat{p}_i, \widehat{c}]_p > 0$.

$$G_1(\widehat{p}_i, \widehat{c}) = \sum_{q=0}^m a_q [\widehat{p}_i, \widehat{c}_q]_p \widehat{c}_q \quad (65)$$

For example, given the coincidence vector in Table II,

$$G_1(\widehat{p}_{32}, \Gamma_2(d)) = f_3 x_{32} x_{20} \widehat{p}_{32} \otimes \widehat{p}_{20} + f_3 x_{32} x_{21} x_{10} \widehat{p}_{32} \otimes \widehat{p}_{10} \\ + f_3 x_{32} x_{21} \widehat{p}_{32} \otimes \widehat{p}_{21} \quad (66)$$

The gate projector may be extended to gate on coincidences, not just single transitions:

$$G_2(\widehat{p}_i, \widehat{p}_l, \widehat{c}) = \sum_{q=0}^m a_q [\widehat{p}_i, \widehat{c}_q]_p [\widehat{p}_l, \widehat{c}_q]_p \widehat{c}_q \quad (67)$$

This can of course easily be extended to higher order coincidences by multiply on a form for each given transition. Let us now consider how detection maps work in the base space on the coincidence algebra bundle.

C. Detection maps in the bundle.

With the detection maps taking a decay vector to a new vector in the path algebra that alters the probabilities to account for detection effects, when we calculate coincidence probabilities in that detection mapped fiber, the probabilities now alter the coincidence multiplication. It is sufficient for now to realize that we may transport a vector in one fiber to another fiber with out having to consider a connection given that a trivial connection can be established on the underlying product vector bundle³. This is important when dealing with multiple detection maps to calculate a given probability vector. Whenever we are multiplying paths together, it must be defined within the fiber of a given point in the base space, i.e, it must be defined for a given decay vector. When taking the coincidence product in a detection mapped fiber, the probabilities that connect the two paths in the product will be the ones defined by the detection map. For transitions of interest, a detection map that accounts for full energy peak efficiencies would be desired. However,

³ A multiplicative connection would allow for the transportation of vectors in a way that scales them to cancel out the change in the algebra. This could be done with the isomorphisms we defined in the global trivialization of the coincidence algebra bundle. However, we require the scaling here as it defines the correct probabilities.

the probabilities that connect the transitions of interest would be from a detection mapped decay vector that accounts for summing out probabilities. Therefore, we wish to calculate summing out probabilities in the fiber at the summing out mapped decay vector, but use vectors transported from other fibers in the base space (the full energy ones). We may therefore have maps between points in the base space, such as $d \mapsto \varphi(d)$, and maps between fibers, $(\varphi(d), \varphi(d)) \mapsto (\varphi'(d), \varphi(d))$. Again, given the trivial product vector bundle underlying the algebra bundle, the manner in which we move vectors around does not matter, it is only the transportation of products that changes. Put more aptly, the detection maps are

not homomorphisms of the coincidence algebra defined on the fibers. When considering the power expansion of detection mapped vectors calculated in the coincidence algebra, we must now distinguish what map is being applied to the vector being multiplied and what detection map is defining the coincidence multiplication. For lifting sections that have been mapped to detection vectors, we denote

$$\tilde{\varphi}(s) := (\varphi(d), s(\varphi(d))). \quad (68)$$

When considering the same problem, of coincidence summing with summing out and summing in, we may use the following power expansion where the fiber is now explicitly stated:

$$\overline{\tilde{\varphi}_h(\tau)}_N^{n_{\varphi_o(\tau)}} = \sum_{k=1}^n \tilde{\varphi}_h(\tau)^{k_{\varphi_o(\tau)}} = \tilde{\varphi}_h(\tau) + \frac{\tilde{\varphi}_h(\tau) \cdot_{\varphi_o(\tau)} \tilde{\varphi}_h(\tau)}{N} + \frac{\tilde{\varphi}_h(\tau) \cdot_{\varphi_o(\tau)} \tilde{\varphi}_h(\tau) \cdot_{\varphi_o(\tau)} \tilde{\varphi}_h(\tau)}{N^2} + \dots \quad (69)$$

Note, we begin the power expansion at $k = 1$ as the coincidence algebra does not have a global identity, only a complete set of idempotents. Regardless though, for the power expansion defining the paths of interest in the probability vector, there should be no $k = 0$ term as we only want path terms; this is why the identity element is subtracted off in Eq. 32. Conversely, in the scalar connection – which is calculated in the path algebra and thus does have a global identity available – the power expansion must start at $k = 0$ to allow for trivial scalar connections between paths that can concatenate. It is also needed for assigning the proper branching ratios. With the detection maps established in the coincidence algebra bundle we may now calculate the probability vector with summing in and summing out maps:

$$\Gamma_1(d) = \tilde{b} \cdot_{\varphi_o(\tau)} \overline{\tilde{\varphi}_h(\tau)}^{n_{\varphi_o(\tau)}} \cdot_{\varphi_o(\tau)} \hat{e}_0. \quad (70)$$

The transitions of interest are given in the center term where summing in is accounted for with potential summing out between disconnected paths. Each path is connected to their possible branching levels above and to the ground state below through the summing out probabilities connecting between. As an example, the probability vector with detection maps for the 4 level decay quiver given by Eq. 70 is shown in Table 1. The basis vectors are separated by the number of full energy gammas they represent. The coefficients are shown in the columns and are distinguished by the branch they decayed from.

To calculate the detected coincidence probability vector $\Gamma_2(d)$ with summing corrections, we must first define a new detection map that allows for the second gamma (with the first being the gate) to be detected—that is, it does not sum with the first gamma. To do this we

Basis Vector	f_3	f_2	f_1
$\hat{e}_3 \otimes \hat{p}_{30}$	$f_3 h_{30}$		
$\hat{e}_3 \otimes \hat{p}_{31} \otimes \hat{e}_0$	$f_3 h_{31} o_{10}$		
$\hat{e}_3 \otimes \hat{p}_{32} \otimes \hat{e}_0$	$f_3 h_{32} o_{20}$		
$\hat{e}_3 \otimes \hat{p}_{21} \otimes \hat{e}_0$	$f_3 o_{32} h_{21} o_{10}$		
$\hat{e}_3 \otimes \hat{p}_{20}$	$f_3 o_{32} h_{20}$		
$\hat{e}_3 \otimes \hat{p}_{10}$	$f_3 (o_{31} + o_{32} o_{21}) h_{10}$		
$\hat{e}_2 \otimes \hat{p}_{20}$		$f_2 h_{20}$	
$\hat{e}_2 \otimes \hat{p}_{10}$		$o_{21} h_{10}$	
$\hat{e}_1 \otimes \hat{p}_{10}$			$f_1 h_{10}$
$\hat{e}_3 \otimes \hat{p}_{31} \otimes \hat{p}_{10}$	$f_3 h_{31} h_{10}/N$		
$\hat{e}_3 \otimes \hat{p}_{32} \otimes \hat{p}_{20}$	$f_3 h_{32} h_{20}/N$		
$\hat{e}_3 \otimes \hat{p}_{32} \otimes \hat{p}_{10}$	$f_3 h_{32} o_{21} h_{10}/N$		
$\hat{e}_3 \otimes \hat{p}_{32} \otimes \hat{p}_{21} \otimes \hat{e}_0$	$f_3 h_{32} h_{21} o_{10}/N$		
$\hat{e}_3 \otimes \hat{p}_{21} \otimes \hat{p}_{10}$	$f_3 o_{32} h_{21} h_{10}/N$		
$\hat{e}_2 \otimes \hat{p}_{21} \otimes \hat{p}_{10}$		$f_2 h_{21} h_{10}/N$	
$\hat{e}_3 \otimes \hat{p}_{32} \otimes \hat{p}_{21} \otimes \hat{p}_{10}$	$f_3 h_{32} h_{21} h_{10}/N^2$		

TABLE III. Probability vector $\Gamma(d)$ coefficients for a 4 level decay quiver with summing out probabilities $o_{ij} = \varphi_o(x_{ij})$ and detected transition $h_{ij} = \varphi_h(x_{ij})$. The basis vectors are separated by the branch they decayed from. The first block contains the basis vectors where 1 gamma is detected at full energy; the second block contains the basis vectors where 2 gammas are detected at full energy; the third block contains the only basis vector where 3 gammas are detected at full energy.

continue to use the isotropic approximation and say the probability that the second gamma does not sum with the first is given by $(N - 1)/N$. We therefore define the gated hit detection map as

$$\varphi_g = \frac{\varphi_h(N - 1)}{N}. \quad (71)$$

Again, with N being the number of detectors. The de-

tected coincidence probability vectors is therefore

$$\Gamma_2(d) = \tilde{b} \cdot \overline{\tilde{\varphi}_h(\tau)}^{n_{\varphi_o(\tau)}} \cdot \overline{\tilde{\varphi}_g(\tau)}^{n_{\varphi_o(\tau)}} \cdot \epsilon_0, \quad (72)$$

where the multiplication $\cdot := \cdot_{\varphi_o(\tau)}$ is over the summing out fiber as is done in Eq.70. However, with higher order coincidence that account for the summing in there is a degeneracy in the basis vectors for different detection scenarios. For example, the basis vector $\widehat{p}_{32} \otimes \widehat{p}_{21} \otimes \widehat{p}_{10}$ could correspond to the summing-in to $\widehat{p}_{31}$ in coincidence with $\widehat{p}_{10}$, or $\widehat{p}_{32}$ in coincidence with the summing-in to $\widehat{p}_{20}$. Calculating the coincidence vector as shown in Eq.72 would provide the probability for both of these distinct detection scenarios onto the same basis vector. To avoid this detection degeneracy (distinct detection events possessing the same basis vector), a gate projector can be applied to the gated hit detection map vector after power expansion.

IV. REMARKS

A. Auxiliary Radiation

A note should be said here on what motivated this framework. As previously mentioned, the transition matrix formalism for summing corrections in nuclear level schemes was initially put together by Semkow. Afterwards, others like [3] extended it by including *auxiliary radiation*; that is, others forms of radiation like X-rays from internal conversion and 511 keV gamma-rays from positron-electron annihilation. It is typical that auxiliary radiation can be added on into the formalism by the use of virtual branches. In this case, branches are added on to the decay level scheme with the additional transitions representing the auxiliary radiation. There are multiple ways to do this corresponding to different types of radiation. For X-rays, [3] showed that virtual branches can be placed between the levels to allow for internal conversion and X-rays. In [6]—predating the work of [2], 511 keV gammas are proposed to be incorporated as pseudo-branches above the level scheme, where internal conversion coefficients of 0.5 double the probability of the emission allowing for the two 511 keV produced in annihilation—though not to be detected simultaneously given a single detector setup. For modern day gamma spectroscopy experiments, like those done with [9], a spherical distribution of multiple detectors is more standard and allows for coincidence measurements. For multiple detector arrangements having two 511 keV gammas below the ground state, as shown in Figure 3, is more suitable. In this manner, we ensure that all decays give the 511 keV radiation from the positron annihilation. The transition matrix formalism can handle the summing out corrections with the 511 virtual branches but is restricted to only allow summing in corrections with the transitions directly connected above the ground state. It is this very need to allow for full energy summing

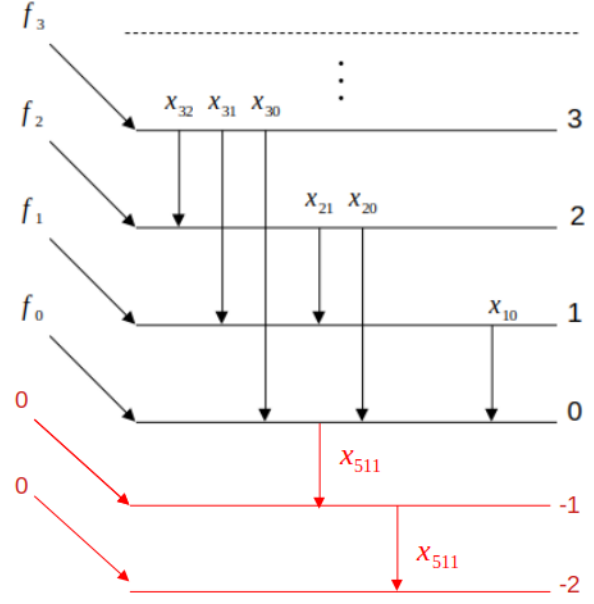


FIG. 3. A general nuclear level scheme with virtual branches below the ground state to allow for 511 keV gammas in the summing formalism.

in corrections with 511 keV gammas in *general* that has motivated this work. However, proposed here is not just a framework that handles that particular case, but handles the combinatorics of all possible coincidences. An example where the 511 keV summing is important is the β^+ decay of ^{22}Mg where summing with 511 keV gammas corrupts one of the large peaks of interest⁴. However, the coincidence algebra not only addresses this by allowing for full energy summing-in between all gammas, but further separates coincidence terms by their branches and makes the extension to gated probabilities trivial.

B. Conclusion

Motivated by the need to extend the generality of the transition matrix formalism traditionally used to model nuclear decay schemes, We have demonstrated how the path algebra of a decay quiver gives the underlying algebra for transitions matrices. This allowed for an extension into a tensor decay space which we have restricted by defining a coincidence vector space upon which we build the coincidence algebra. The coincidence algebra allows for a more general algebra for calculating coincidence probabilities as it allows for multiplication between

⁴ The precision measurement of the superallowed β^+ decay of ^{22}Mg —a measurement that contributes to the V_{ud} value in the CKM matrix—requires a precise measurement of this corrupted peak.

non connected paths in the decay quiver and in addition separates different branching terms. With the path algebra being the space with which nuclear levels schemes live, we have define an algebra bundle over the path algebra where each decay vector defines the structure constants on the coincidence algebra at the fiber. We have

showed, at the very least that this algebra allows for additional coincidence probability terms to be calculated. We propose, however, that this method of taking fiber bundles of quivers, or potentially even quiver representations, may be a fruitful method for modeling all nuclear levels schemes.

* A footnote to the article title

† Also at Physics Department, University of Guelph.

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